

浙江大学 2025–2026 学年春夏季学期

《数据库系统》课程课堂测试二

(Quiz 2 for Database Systems)

考生姓名: _____ 学号: _____ 专业: _____ 得分: _____

Consider the following relation schemas and then answer the subsequent problems. Note that the key attributes in the relation schemas are underlined.

Movie(title, type, director)

Comment(title, user_name, grade)

Problem 1. Relational algebra

(1) Find all the movie titles that are directed by “Yimou Zhang” and exist the comment grade of greater than or equal to 4. (16 points)

$\Pi_{title}(\sigma_{director="Yimou Zhang"}(Movie) \bowtie \sigma_{grade \geq 4}(Comment))$

(2) Find the titles of movies which are commented by “San Zhang”. (17 points)

$\Pi_{title}(\sigma_{user_name="San Zhang"}(Comment))$

(3) Find the lowest grade in *Comment* Table. (17 points)

Method 1: $\Pi_{grade}(Comment) - \Pi_{grade}(\sigma_{Comment.grade > cmt2.grade}(Comment \times (\rho_{cmt2}(Comment))))$

Method 2: $g_{min(grade)}(Comment);$

(Different answer is OK if it is correct.)

Problem 2. Write SQL statement for the following queries.

(1) Find all the movie titles and types with their average grade. (16 points)

(Different answer is OK if it is correct.)

SELECT title, type, avg(grade)

FROM Movie M, Comment C

WHERE M.title=C.title

GROUP BY title

(2) Find the directors of the movies that have the lowest average grade. (17 points)

(Different answer is OK if it is correct.)

```

SELECT director
FROM Movie M, Comment C
WHERE M.title=C.title
GROUP BY title
HAVING avg(grade)<=all (SELECT avg(grade)
                        FROM Movie, Comment
                        WHERE Movie.title=Comment.title
                        GROUP BY title)

```

**(3) Find all the movie titles where every user gives higher grade than movie “the avenger”.
(17 points)**

(Different answer is OK if it is correct.)

```

SELECT title from Movie
EXCEPT
SELECT title from movie
WHERE EXISTS (SELECT *
              FROM Comment A, Comment B
              WHERE A.title=movie.title and A.user_name = B.user_name
              AND B.title='the avenger'
              AND A.grade <=B.grade)

```

```

Student(SID, name)
Course(CID,name,Teacher)
SC(SID,CID)

```

检索学过高老师主讲的所有课程的同学姓名.

```

SELECT name FROM Student
WHERE NOT EXISTS                                     // 不存在
    SELECT * FROM course                             // 有一门高老师的课
WHERE course.Teacher='高云君' AND NOT EXISTS         // 该同学没学过
    SELECT * FROM SC
WHERE Student.SID=SC.SID AND Course.CID=SC.CID

```